



Explainable AI for government: Does the type of explanation matter to the accuracy, fairness, and trustworthiness of an algorithmic decision as perceived by those who are affected?

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ABSTRACT

Amidst concerns over biased and misguided government decisions arrived at through algorithmic treatment, it is important for members of society to be able to perceive that public authorities are making fair, accurate, and trustworthy decisions. Inspired in part by equity and procedural justice theories and by theories of attitudes towards technologies, we posited that the perception of these attributes of decisions is influenced by the type of explanation offered, which can be input-based, group-based, case-based, or counterfactual. We tested our hypotheses with two studies, each of which involved a pre-registered online survey experiment conducted in December 2022. In both studies, the subjects ($N = 1200$) were officers in high positions at stock companies registered in Japan, who were presented with a scenario consisting of an algorithmic decision made by a public authority: a ministry's decision to reject a grant application from their company (Study 1) and a tax authority's decision to select their company for an on-site tax inspection (Study 2). The studies revealed that offering the subjects some type of explanation had a positive effect on their attitude towards a decision, to various extents, although the detailed results of the two studies are not robust. These findings call for a nuanced inquiry, both in research and practice, into how to best design explanations of algorithmic decisions from societal and human-centric perspectives in different decision-making contexts.

1. Introduction

The public sector is becoming increasingly reliant on decision-making driven by machine learning (ML) algorithms (de Sousa et al., 2019; Janssen et al., 2022; van Noordt & Misuraca, 2022; Wang et al., 2023); however, concerns have been raised over the discriminatory treatment of members of society caused by biased input data (Anik & Bunt, 2021; The Defender of Rights, Republic of France, 2020; Dodge et al., 2019; Weyerer & Langer, 2019). Moreover, current ML systems are based on correlation rather than causation (Arrieta et al., 2020), which could lead to misguided decisions (Hao, 2019). ML also involves risks because it is unable to deal with unusual cases (Janssen et al., 2022). Along with these flaws, decisions with ML algorithmic treatment (hereafter simply "algorithmic decisions") are neither transparent (Lepri et al., 2018), accountable (Burrell, 2016; de Laat, 2018), nor contestable (Wachter et al., 2018), due to the opacity of the systems.

Against this backdrop, explainable artificial intelligence (XAI) is considered a remedy for opaque algorithmic decisions (de Bruijn et al., 2022; Janssen et al., 2022). It is an area of artificial intelligence (AI) research that emerged as early as the 1970s (Buchanan & Shortliffe, 1984). An XAI system is designed to make an AI system's behavior "more intelligible to humans by providing explanations" (Gunning et al., 2019, p. 1) of the underlying causes for its or another agent's decisions and behaviors (Miller, 2019). By doing so, it aims to enable "the human (or end user) to appropriately calibrate trust and reliance, to detect potential errors in machine reasoning, and to help audit the technology" (Miller et al., 2022, para. 2). XAI has received attention through the passage of laws and regulations designed to protect an individual's right to an explanation for an administrative decision based on algorithmic treatment (Brkan & Bonnet, 2020; Haque et al., 2023).

Given the salience of XAI, we sought to examine whether the type of explanation offered via XAI makes a difference to the fairness, accuracy,

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and trustworthiness of a decision, as perceived by those who are affected by it (hereafter referred to as “affectees”). By doing so, we hoped to advance research that can ultimately guide how to best design explanations from societal and human-centric perspectives.

We chose fairness, accuracy, and trustworthiness because, in light of the abovementioned issues with biased and erroneous algorithmic decisions, combined with the coercive societal implications of government decisions (Aoki, Tay, & Yarime, 2024; Rainey et al., 1976), it is important to ensure that members of society be able to recognize that public authorities are making fair, accurate, and trustworthy decisions. Such positive recognition might also play an important role in securing overall trust in government.

Previously, Cramer et al. (2008) and Grimmelikhuijsen (2022) arrived at mixed findings with regard to how offering an explanation can influence trust in a recommender system. Kizilcec (2016) found that the level of detail in an explanation had an effect on a compound measure of trust, based on ratings of users’ understanding of, and trust in, an algorithmic peer-assessment system and its fairness and accuracy. Williamson and Kizilcec (2021) revealed that the level of visualization of an explanation also mattered to the perceived accuracy of, and trust in, the algorithm behind a learning application. Cheng et al. (2019) posited that trust in profiling algorithms might depend on the static or interactive nature of the explanation interface, although they did not find support for this hypothesis.

Our research is more in line with studies by Binns et al. (2018) and Dodge et al. (2019), in that, like them, we focused on types of explanations. Using UK data, Binns et al. (2018) considered four types of explanations, categorized as input influence-based, demographic-based, case-based, and sensitivity-based. One finding relevant to our study was that the subjects’ perception of the fairness of a loan assessment was stronger when they received a sensitivity-based explanation rather than a case-based explanation. Dodge et al. (2019) noted that the four types of explanations used by Binns et al. (2018) “represent a set of common approaches in recent XAI work” (para. 1 in 3.2) and found that they mattered to the perceived fairness of ML systems in the context of US criminal justice.

Inspired by studies like that of Binns et al. (2018) in the XAI literature, we considered four types of explanations, which we called “input-based,” “group-based,” “case-based,” and “counterfactual,” and hypothesized that the offering of an explanation (its status), and the type of explanation, matter to the fairness, accuracy, and trustworthiness of a decision as perceived by its affectees. To test our hypotheses, we conducted two studies (Study 1 and Study 2), each involving a pre-registered, online survey experiment. In both studies, the subjects ($N = 1200$) were officers in high positions at stock companies registered in Japan, who were presented with a scenario consisting of a ministry’s decision to reject a grant application from their company (Study 1) or a scenario involving a tax authority’s decision to select their company for an on-site tax inspection (Study 2). We designed these decisions to simulate common decisions that public authorities in Japan actually make, so that we expected our subjects to be able to relate to these decisions as realistic.

While XAI has been seen as an area mainly for “developers in computer science” (Meske et al., 2022, p. 53), we conducted this research from societal and human-centric perspectives. While the number of studies written from the latter perspective has been growing in recent years (Haque et al., 2023), research on the effects of the characteristics of explanations in general, and in particular on types of explanations, as in Binns et al. (2018) and Dodge et al. (2019), nevertheless remains rare. We complement their pioneering research by presenting an alternative experimental approach, in hopes of better isolating the effects of explanation types, and contribute empirical findings for decision domains that differ from theirs and come from a different place – Japan.

The rest of this paper proceeds as follows. The next section discusses the types of explanations we identified based on the current XAI literature. The third section develops our hypotheses. The fourth explains

the empirical approaches underlying the two studies. The fifth section presents the results from the two studies, and the sixth offers a discussion of the results and findings. The seventh section concludes the paper.

2. The types of explanations considered in our studies

In our study context, an *explanation* was considered a causal history of an explanandum event. For someone (an *explainer*) to engage in an act of explaining, they must be “in possession of some information about the causal history of some event [and] ... [try] to convey it to someone else” (Lewis, 1986, pp. 217–218); the mere availability of the information is not an explanation, because an explanation should involve the act of conveying the information. If we apply this definition of *explanation* to algorithmic decisions by public authorities, the explanandum event is the decision made solely or predominantly by ML algorithms, while the explanation concerns why and how the decision was reached.

The *explainer* can be another AI agent, XAI that generates an explanation, or an administrative officer who explains the XAI’s explanation to the *explainee*. Hereafter, we use the term *explainee* in contradistinction to the terms *data subject* and *affectee*. While the term *data subject* has been used to refer to a natural person (e.g., Regulation (EU) 2016/679, 2024; Wachter et al., 2018), we define it broadly in the context of our study to include other entities whose information is collected and assessed by an AI system, such as companies. On the other hand, we use the term *affectee* to denote an individual or an organization that is affected directly by a decision, as well as an individual who has a stake in the organization. An affectee may be considered an explainee only if they are the recipient of an explanation.

The question of how an administrative decision ought to be explained arises because there is no single way to offer an explanation. A single explanation is not expected to capture the universe of all causes of the explanandum and inevitably falls short of wholeness (Lewis, 1986; Miller, 2019). As a result, any explanation is selective, in that it is based on causal historical information selected by the explainer. Explanations can differ in form and length and in their degree of abstraction or specificity (Lewis, 1986). Aspects of explanations such as these can impact individuals’ attitudes towards a decision that affects them. Our studies focused on explanations that differ with respect to the kind of information they present. We categorized the explanations according to their type, inspired by Binns et al. (2018) and Dodge et al. (2019), as well as by other XAI literature, cited below.

2.1. Input-based explanations

An *input-based explanation* – what Binns et al. (2018) call an *input influence-based explanation* – focuses on the internal logic that yields the output; for instance, it provides information on input variables such as race and gender and describes how much each contributed to the output (Binns et al., 2018). This type of explanation is offered by XAI which works “by analyzing various inputs used by a decision-making algorithm, measuring the impact of each of the inputs individually and in groups, and finally reporting the set of inputs that had the biggest impact on the final decision” (Hosanagar & Jair, 2018, para. 14). Binns et al. (2018) presented an example of an input influence-based explanation for an algorithmic decision on the amount of a car insurance premium, based on the probability that the applicant would be involved in an accident. The inputs were age, driving experience, the level of adherence to speed limits, the number of trips taken at night, and miles per month. In the explanation, numbers of positive (+) or negative (–) signs were used to indicate the degree of influence of these inputs on the applicant’s chance of being in an accident.

2.2. Group-based explanations

A *group-based explanation* involves reporting outcome distributions for other cases in the AI’s training dataset that involve the same

demographic or attribute categories as those of the data subject in question. This term is our re-naming of what Binns et al. (2018) called *demographic explanations* in the context of the car insurance premium decisions mentioned above, such as “29% of female drivers qualified for the cheapest tier” of the premium and “31% of drivers in your age group [30-39] qualified for the cheapest tier” (p. 6). We chose the term *group-based* instead of *demographic* because AI decisions are not always made for individuals; they can be made for organizations, such as the companies in our experiments, to which the term *demographic* does not apply. Because the examples given by Binns et al. (2018) might sound as though they concern the percentage distribution of algorithmic decision outcomes, it is worth repeating that the group-based explanation in our studies concerns the percentage distribution of outcomes in the AI’s training data.

2.3. Case-based explanations

AI recommender systems learn to perform by training with data from numerous past cases. XAI based on *case-based reasoning* is “designed to find the most similar cases of a query case in the database” and display them to the user, who can then compare them with their own case and assess the decision (Li et al., 2022, p. 2259). This type of explanation has been used for AI systems that predict house prices (Keane and Kenny, 2019), assess financial risk (Li et al., 2022), and make medical recommendations (Doyle & Cunningham, 2006). Binns et al. (2018, p. 6) offer an example of this type of explanation using the car insurance premium scenario above, which states that “[t]his decision was based on thousands of similar cases from the past” and that “a similar case to yours is a previous customer, Claire,” followed by a description of Claire’s case.

2.4. Counterfactual explanations

Wachter et al. (2018) advocate the use of *counterfactuals* as explanations for automated decisions based on algorithmic treatment. To them, counterfactuals concern “how the world would have to be different for a desirable outcome to occur,” for example, “You were denied a loan because your annual income was £30,000. If your income had been £45,000, you would have been offered a loan” (p. 844). Explanations like these are what Binns et al. (2018) call *sensitivity-based explanations*; they explain how much of an input variable would have had to change to produce a different outcome. The example they offer is the following: “If 10% or less of your driving took place at night, you would have qualified for the cheapest tier [of the car insurance premium]” (p. 6). Here, we choose to use the term “counterfactual” over “sensitivity-based,” because the former is more fit to describe an explanation given by a public authority to the lay public, as opposed to “sensitivity,” which has a strong connotation of mathematical modeling, such as in “sensitivity” analysis. The crux of a counterfactual explanation is “the smallest change to the world that can be made to obtain a desirable outcome” (Wachter et al., 2018, p. 845). Unlike the other types of explanations, counterfactual explanations are not about the internal logic or the nature of the training data; they relate to the cause of the explanandum event that occurred, relative to eventualities that did not (Miller, 2019).

3. Hypotheses

We next consider the reasons for why the availability of an explanation and its type make a difference in affectees’ attitudes towards an algorithmic administrative decision. Following previous studies (e.g., Binns et al., 2018; Kizilcec, 2016), we focus on decisions made against the interests of the affectee, because affectees, as a rule, want an explanation when a decision is upsetting, not when a decision is favorable. For *attitudes*, we selected three outcomes, namely, the perceived fairness, accuracy, and trustworthiness of the decision, because it is important for decisions made by public authorities to be seen to possess

these attributes, as mentioned earlier.

3.1. Effects on perceived fairness

According to psychological theories, factors other than an outcome itself can critically influence people’s attitudes towards the outcome. For instance, the equity theory (Adams, 1965) posits that people assess the fairness of outcomes in reference to the equity that exists when the ratio of outcomes (e.g., rewards like pay and job status in a work setting) to inputs (e.g., investments like effort, skill, seniority, and demographic attributes) for themselves and the ratio of a reference person or persons are equal. The comparison of these two ratios influences their perception of the outcomes they have received.

While the equity theory concerns distributive justice in the aforementioned equity sense, studies on the psychology of procedural justice, put forward most notably by Thibaut and Walker (1975), suggest that the perceived fairness of the process or procedure regulating an outcome makes a difference in one’s attitude towards the outcome (Folger & Greenberg, 1985; Greenberg & Folger, 1983). Leventhal (1977) proposed that for a procedure to be seen as fair, it must be (i) consistent across persons and times, (ii) free of bias, (iii) accurate, based on “good information and informed opinion” (p. 25), (iv) correctable, in that “opportunities must exist to modify and reverse decisions” (p. 27), (v) representative, in that the process should reflect the concerns and values of those who are affected by it, and (vi) ethical. Bies and Moag (1986) added the concept of interactional justice, proposing that procedural justice depends in part on how truthful and respectful interpersonal communication is during the judicial process.

Inspired by the theories cited above, we posited that whether an explanation about a negative algorithmic administrative decision is available (its status) makes a difference in affectees’ perceived fairness of the decision process, and thus, of the decision itself. Paramount is the fact that any explanation is better than no explanation at all, because affectees feel that they are being treated with respect (Simmons, 2018), which is consistent with Bies and Moag’s theory of interactional justice, mentioned above. Moreover, the fact that an explanation is provided enhances the perception that the process is transparent and likely to be free of bias, which meets one of the fairness criteria proposed by Leventhal. As a result, explainees believe that the process and the resulting decision are fairer than they would seem without an explanation. This logic is somewhat supported by the empirical finding that offering an explanation mitigates a negative response to a negative outcome (e.g., Greenberg, 1990).

Furthermore, we posited that different types of explanations offer different reference points for justice assessment; as a result, the type of explanation an explainer chooses to give makes a difference in the explainee’s perceived fairness of the decision. For instance, input-based explanations afford an assessment of procedural fairness, in respect to the criteria used and how much each criterion contributes to the decision. Group-based and case-based explanations offer different points of comparison for the assessment of distributive justice. Explainees may perceive the decision to be fairer if the explanation offers a group as a reference, as in a group-based explanation, rather than referring to a single case from the training data.

Explanations also differ in scope. They are global when they intend to help the explainee understand the model as a whole and local when they provide an explanation of individual data instances (Das & Rad, 2020; Dodge et al., 2019). In other words, a global explanation is model-specific, while a local explanation is instance-specific (Das & Rad, 2020). The examples of case-based and counterfactual explanations provided above are likely to sound local and instance-specific rather than global. Input-based and group-based explanations are supposed to be global by design; however, explainees will not take them that way if they understand the input contributions to be specific to the particular decision at hand, without recognizing that they apply to the whole model, and if they use the information on the outcome distributions in the training

data only to understand their own outcome and relative standing. In short, perceptions matter. In general, providing an explanation perceived to be global might contribute more to perceived fairness than an explanation perceived to be instance-specific, because the information in the former is universal and speaks to the functionality of the entire system.

Counterfactuals contribute in a unique way to the perception of procedural fairness from a due process point of view. Unlike explanations that focus on helping the explainee to understand the internal workings of an AI system or its training data or the nature of the training data, counterfactual explanations provide explainees with “meaningful explanations to understand a given decision, grounds to contest it, and advice on how the data subject can change his or her behaviour or situation to possibly receive a desired decision” (Wachter et al., 2018, p. 844). Therefore, counterfactuals help explainees to challenge and take action against automated decisions. This possibility enhances the perceived fairness of the entire process by fulfilling Leventhal’s (1977) corrective dimension of a fair process.

Taken all together, equity and procedural justice theories help us to think through how offering an explanation and how the type of explanation might impact affectees’ assessments of distributive and procedural justice and, thus, their perceived fairness of a decision made by administrative authorities. Consequently, we proposed the following hypotheses, H1(i) and H1(ii):

H1. An affectee’s perceived fairness of an adverse algorithmic administrative decision depends (i) on whether an explanation for the decision is provided and (ii) on the type of explanation provided.

3.2. Effects on perceived accuracy

Even without explanations, individuals might perceive automated decisions as accurate, due to humans’ general positive bias towards expert systems (Dijkstra, 1999; Dzindolet et al., 2003). However, the opposite is possible if they have developed “algorithmic aversion” – a psychological phenomenon whereby “people will generally choose predictions made by human beings over those made by computers, even after they have seen evidence that the computer predictions are more accurate” (Simmons, 2018, pp. 1090–1091). Such an aversion develops when individuals overreact to a mistake made by an algorithm (Dietvorst et al., 2015). In either case, an explanation might enhance perceived accuracy because it can help system builders, if not affectees themselves, review the contents of a system’s knowledge base and its lines of reasoning, so they can monitor system activities and identify and remove errors (debug the system) (Adadi & Berrada, 2018). As a result, an explanation reinforces the belief that a system’s decisions are subject to correction and, therefore, accurate.

The effect of an explanation on the perceived accuracy of a decision might also depend on the type of explanation. It is possible that explanations perceived to be global in scope have a more positive impact on perceived accuracy than explanations perceived to be instance-specific, because the former can help with the monitoring of a system’s line of reasoning and enable designers to check for biases in training data or a system’s knowledge base, assuming that the explainees are aware of these functionalities. On the other hand, instance-specific explanations, such as case-based and counterfactual explanations, might be seen to merely justify a particular decision rather than explain the model or the training data underlying the model, and so do not help subjects or experts locate errors in the system as a whole. In light of these considerations, we tested the following hypotheses, H2(i) and H2(ii):

H2. An affectee’s perceived accuracy of an adverse algorithmic administrative decision depends (i) on whether an explanation for the decision is provided and (ii) on the type of explanation provided.

3.3. Effects on perceived trustworthiness

In the context of the criminal justice system, Simmons (2018) expresses concern that the use of predictive algorithms is likely to harm procedural justice and, hence, the trustworthiness of the system. This occurs due to a reduction of interpersonal respect paid by authorities to defendants, the loss of a personal connection between defendants and the authorities, and a loss of process control on the part of defendants. Simmons (2018) argues that harm to perceived procedural justice negatively affects the trustworthiness of the system and recommends that authorities “explain the details of the algorithm to the defendants rather than just saying that a machine mandated the decision” (p. 1081). The importance of explaining decisions for procedural justice, and thereby reinforcing the trustworthiness of the decisions, is likely to apply beyond the context of criminal justice. The importance of explaining is also put forward by psycho-ergonomics theory, which views people’s understanding of the processes behind automated operations as one of the sources of human trust in automation (Lee & Moray, 1992; Lee & See, 2004). The four types of explanations highlight different types of information regarding the decision-making process; however, global explanations concerning the model as a whole might help explainees comprehend the process more than local explanations and, thus, contribute more to perceived procedural justice. As a result, we proposed the following hypotheses, H3(i) and H3(ii):

H3. An affectee’s perceived trustworthiness of an adverse algorithmic administrative decision depends (i) on whether an explanation for the decision is provided and (ii) on the type of explanation provided.

4. Methods

4.1. Study design

To test our hypotheses, we conducted two survey experiments online (Study 1 and Study 2), which were approved by our university’s Research Ethics Committee and pre-registered via the Open Science Framework. The study pre-registrations, along with other study materials (data, codes, and the full questionnaires in Japanese and their English translations) can be found at <https://osf.io/zr8gn/> for Study 1 and at <https://osf.io/jr96f/> for Study 2.

The two experiments involved scenarios, i.e., descriptions of hypothetical events. The scenario in Study 1 concerned a decision made by the Ministry of Economy, Trade and Industry to reject a business-development grant application submitted by the respondent’s company. In Study 2, the scenario involved a decision by tax authorities to conduct an on-site tax inspection of the respondent’s company. The use of scenarios enabled us to test the effects of different types of explanations that have yet to be deployed by public authorities and to manipulate the status and type of explanation, as elaborated later in Section 4.3.

Studies 1 and 2 were similar in that they both consisted of an online survey experiment with a between-subject design and involved one factor (i.e., no explanation or one of four types of explanation regarding an AI-assisted decision); in addition, the outcome measures in both were the perceived fairness, accuracy, and trustworthiness of the decision (to be explained below). Figs. 1 and 2 show the experimental procedures for Studies 1 and 2, respectively.

4.2. Subjects

Both Studies 1 and 2 targeted subjects aged 18 or above who were *yakuin* (officers – namely, directors, accounting advisors, and company auditors) at *kabushikigaisha* (stock companies) registered in Japan, who would be the affectees, along with their companies, of administrative decisions like those in the scenarios.

We recruited these subjects on December 13, 2022. Based on an a

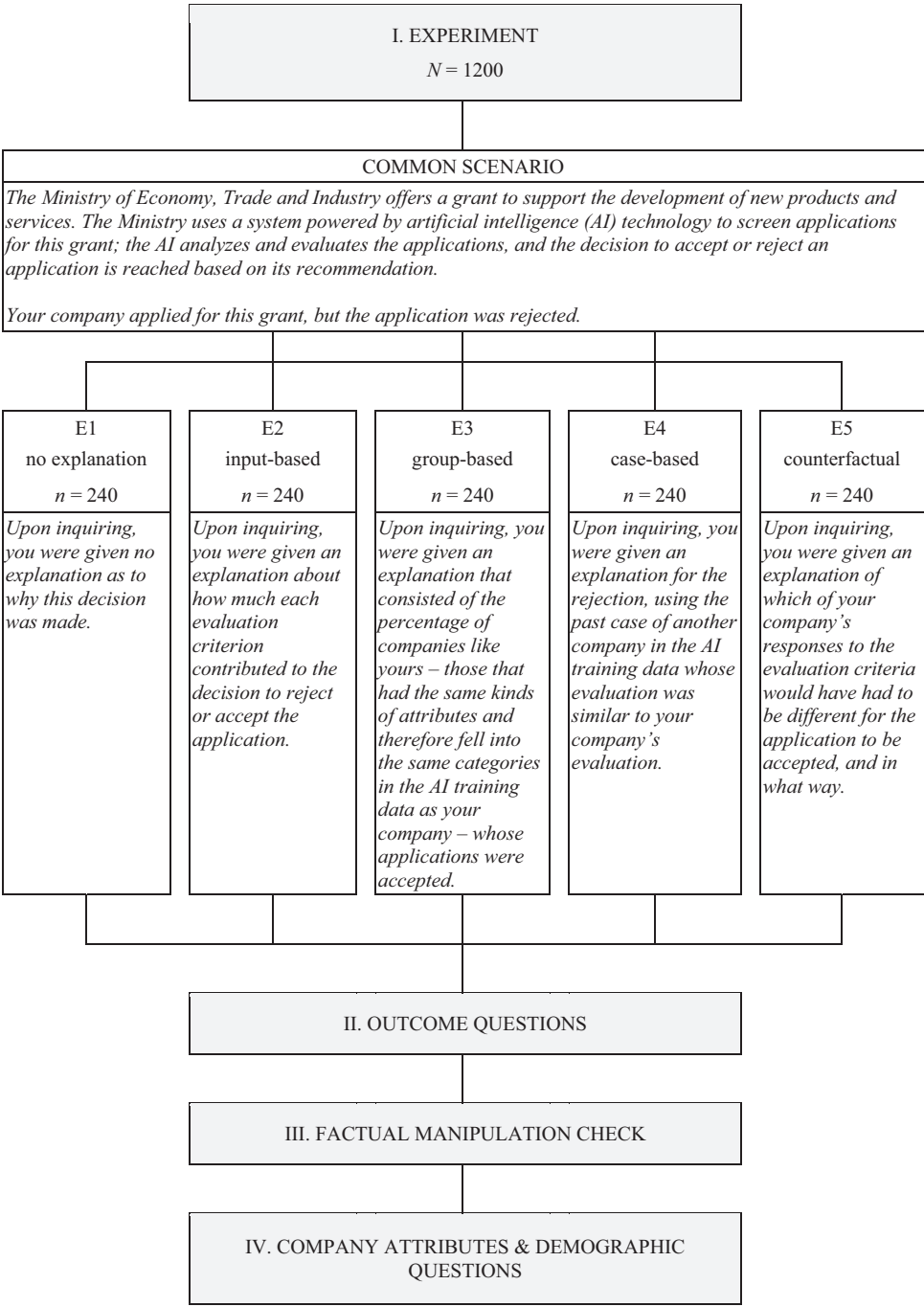


Fig. 1. Experimental procedure and scenario descriptions (English translations) in Study 1.

priori power analysis using the software program G*Power, we requested 1200 complete survey forms for each study; our goal in designating this sample size was to obtain a power of 0.80 to detect an effect size, f , of 0.10 at the 0.05 alpha error probability. Rakuten Insight, Inc., distributed an invitation to two groups, one for Study 1 and the other for Study 2, randomly selected from a panel of registrants who had earlier indicated that they were *yakuin* or in high corporate positions. We used screening questions to identify willing invitees who were still *yakuin* at the time of the survey and who met the age criterion. Those who consented to participate onscreen proceeded to the survey and, as compensation for completing it, earned Rakuten Points, which can be used for future purchases at Rakuten's online shopping malls. At the recruitment stage, randomly selected invitees for Studies 1 and 2 answered the same screening questions and read the same informed

consent statements with no foreknowledge of the decision scenarios in the questionnaire (i.e., grant application or tax inspection).

The two sample groups were similar, as shown in the descriptive statistics in Supplementary Material. The subjects were 24 to 80 years old (Study 1 [S1]: mean [M] = 57.59, standard deviation [SD] = 8.76; Study 2 [S2]: M = 57.32, SD = 8.77), and the majority had a college degree (S1: 68.83 %, S2: 68.33 %). About half (S1: 51.17 %, S2: 51.92 %) earned an annual income of eight million yen or more. Most of the subjects were men (S1: 95.00 %, S2: 94.42 %), which was not surprising, given the fact that as of 2021 only 7.5 % of the *yakuin* at listed companies in Japan were female (Gender Equality Bureau, Cabinet Office, 2022). The subjects' companies operated in various business areas and differed substantially in their number of regular employees, amount of sales, and amount of initial capital.

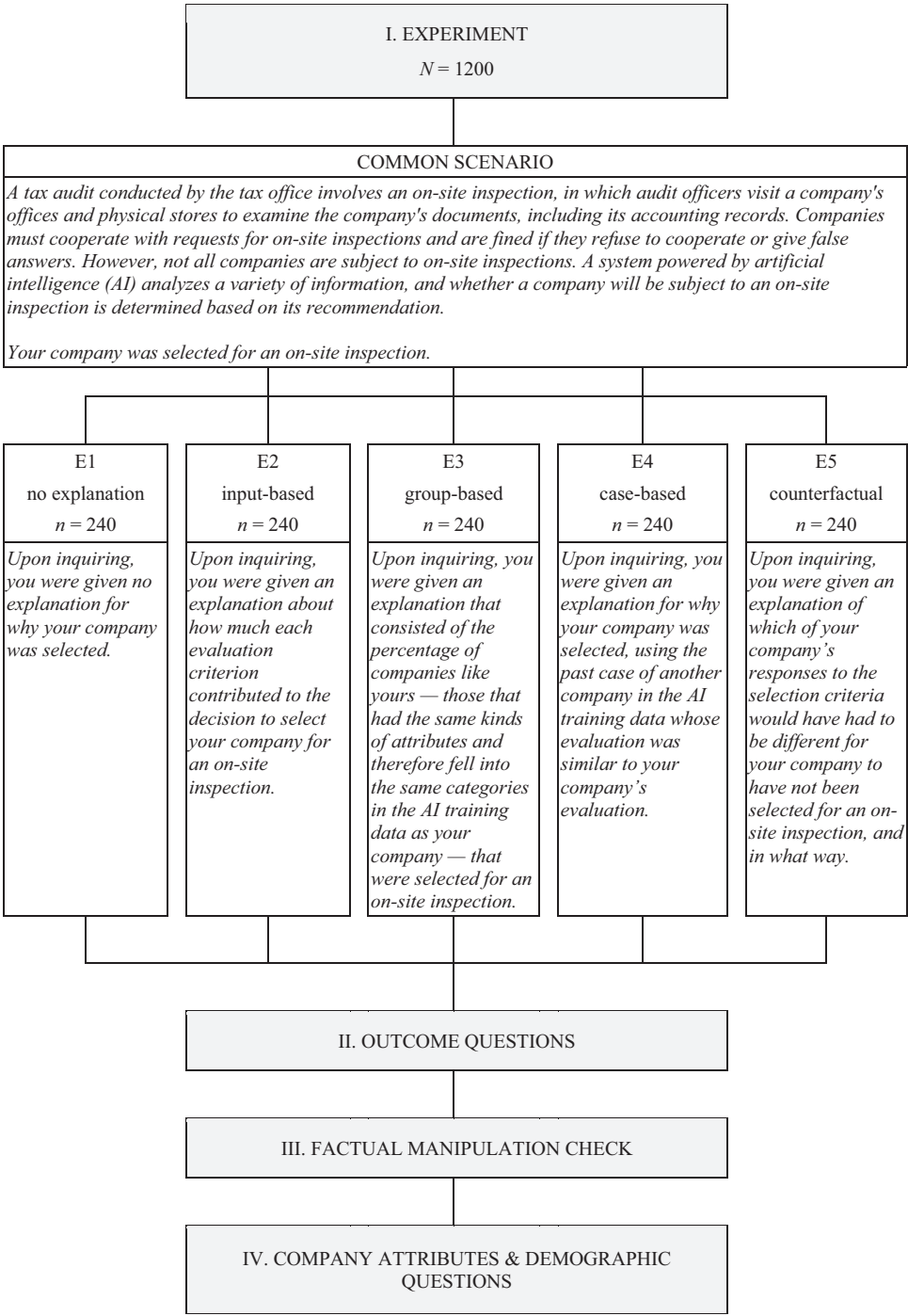


Fig. 2. Experimental procedure and scenario descriptions (English translations) in Study 2.

4.3. Experimental manipulation

In Study 1, the subjects were asked to read a scenario regarding a business-development grant application, which informed them that an AI-assisted judgment had been made against their company and that their application had been rejected. In Study 2, the subjects were informed that it had been decided that their company would be scheduled for an on-site tax inspection. In both studies, we varied the scenarios in regard to the availability of an explanation and the type of explanation. These variations were: no explanation (E1), an input-based explanation (E2), a group-based explanation (E3), a case-based explanation (E4), and a counterfactual explanation (E5). Each subject was randomly assigned to receive a version of the survey with one of these

variations, which was done by computer to achieve double-blind randomization.

When preparing the scenarios, we took care to ensure that they differed only in the textual description of the status and type of explanation provided. Unlike [Binns et al. \(2018\)](#), which used emulations of actual explanations, or [Dodge et al. \(2019\)](#), which used explanations generated using real data, we intentionally avoided presenting realistic or real numerical values or information. If we had done so, we would have had to present different kinds of information for E2 to E5, such as the magnitude of the contribution of certain variables involved in the decision and the acceptance rates for certain groups of companies. Had this been the case, we might not have been able to isolate the effects of the different types of explanations from the effects of the numerical

information. For the same reason, we did not use visual or color representations of explanations, because studies have found that visual representations of explanations make a difference in attitudes towards an outcome (Williamson & Kizilcec, 2021), and the visuals would have had to differ across the different types of explanations.

4.4. Measures of attitudes

In both studies, after reading the scenarios, the subjects were asked to express the extent to which they agreed with the statements: (1) that the decision was fair, (2) that the decision was accurate, and (3) that the decision was trustworthy, using a scale of 0–10, 0 being “do not agree at all” and 10 being “strongly agree” (hereafter referred to as “outcome questions”). Their responses to these questions were our outcome measures, and we randomized the order of the questions to minimize order bias in our analyses.

4.5. Randomization checks

The randomization of experimental conditions (E1–E5), if successfully achieved, eliminates the influence of extraneous variables on the effects of the experimental conditions and thereby eliminates the need to control for variables. However, it can be hard to achieve randomization in an online survey experiment if some subjects withdraw from the survey after answering the outcome questions and if their reasons for doing so are systematically related to the experimental manipulation they encountered. Although such systematic withdrawal was unlikely in both Studies 1 and 2, we nevertheless performed a randomization check (see Tables S-2 and S-3 for the results of the multinomial logistic regressions). For both studies, the results show that there were no notable systematic differences in the subjects’ key demographic categories and their company profiles across the experimental conditions.

4.6. Analytical approach

Our analyses included a one-way ANOVA, in which a $p < .05$ threshold was used to reject the null hypothesis (that there would be no difference in the means of the outcome measures – namely, in the subjects’ perceived fairness, accuracy, and trustworthiness of the AI-assisted decisions) across the five experimental groups (E1 to E5). We also conducted Tukey pairwise multiple comparison tests to determine which pairs of experimental groups were statistically significantly different from each another.

In these analyses, we applied the intention-to-treat (ITT) principle to estimate the effects of the experimental conditions on the outcome measures. In each study, the factual manipulation-check question, included after the outcome questions, asked the subjects which statement they had read in the scenario on the previous page, with a list of options, from E1 at the top to E5 and “do not know” at the bottom. We included this question to enable us to investigate possible reasons for any insignificant results or small effects. Following the ITT principle, it was not our intention to eliminate subjects who failed this question. Doing so would have interfered with the randomization (Montgomery et al., 2018) and would have unnecessarily eliminated responses that represent valid psychological and attitudinal reactions arising from the fact that the manipulated condition was simply not important enough to the subjects for them to remember it.

5. Results

5.1. Main results from the ANOVA and pairwise-comparison tests

Reported below are the main results for the two studies, in sequence.

5.1.1. Study 1: grant applications

The ANOVA test indicated that for all the outcome measures there

was a statistically significant difference in means, with small effect sizes, for fairness ($F_{4,1195} = 7.96, p < .001, \eta^2 = 0.026$), accuracy ($F_{4,1195} = 8.52, p < .001, \eta^2 = 0.028$), and trustworthiness ($F_{4,1195} = 7.79, p < .001, \eta^2 = 0.025$). Based on the Tukey pairwise comparison tests, Fig. 3 compares the mean scores of the outcomes for the three attitudes across the experimental groups and indicates pairs with statistically significant differences. As for fairness, the results show that E2 ($M = 4.84, SD = 2.04$), E4 ($M = 4.85, SD = 2.17$), and E5 ($M = 4.84, SD = 2.28$) score higher than E1 (no explanation) ($M = 3.93, SD = 2.48$) ($p < .001$ for all pairs). The same can be said for perceived accuracy, with E2 ($M = 4.65, SD = 2.03$), E4 ($M = 4.63, SD = 2.19$), and E5 ($M = 4.63, SD = 2.23$) scoring higher than E1 ($M = 3.70, SD = 2.38$) ($p < .001$ for all pairs), as well as for perceived trustworthiness, with E2 ($M = 4.58, SD = 2.04$), E4 ($M = 4.61, SD = 2.16$), and E5 ($M = 4.63, SD = 2.22$) scoring higher than E1 ($M = 3.75, SD = 2.43$) ($p < .001$ for all pairs). However, when explanations were provided, none of the possible pairs of explanations involving E2 to E5 was statistically significantly different at our $p < .05$ threshold on all outcome measures, except the pair E3 ($M = 4.06, SD = 2.25$) and E5 for perceived trustworthiness, where E5 scored higher than E3 by 0.57 ($p = .042$).

5.1.2. Study 2: on-site tax inspection

In Study 2, the results from the one-way ANOVA suggest that we can reject the null hypothesis (that the means would be the same across the five experimental groups), with effect sizes smaller than those in Study 1, for accuracy ($F_{4,1195} = 3.44, p = .008, \eta^2 = 0.011$) and for trustworthiness ($F_{4,1195} = 3.24, p = .012, \eta^2 = 0.011$), but we were not able to reject the null hypothesis for fairness ($p = .111$). Based on the pairwise comparison tests, Fig. 4 compares the mean scores of the outcomes for the three attitudes across the experimental groups. The pairs E1 ($M = 4.18, SD = 2.11$) and E2 ($M = 4.75, SD = 1.92$) ($p = .021$), and E1 and E3 ($M = 4.81, SD = 2.02$) are statistically significantly different for accuracy ($p = .008$), as are the pair E1 ($M = 4.17, SD = 2.17$) and E3 ($M = 4.82, SD = 2.07$) for trustworthiness ($p = .006$).

5.2. Testing with covariates

As noted earlier, our randomization check showed that there were no notable systematic differences in the subjects’ key demographic categories and their company profiles across the experimental conditions. Nevertheless, we performed ANOVA and pairwise comparisons while controlling for these variables. The results are presented and discussed in detail in the Supplementary Material. They remain largely similar to the main results.

6. Discussion

In Study 1, we found that providing any type of explanation except E3 made a statistically significant and positive impact on all of the attitudinal measures. As we hypothesized, the positive impact of explanations on fairness and trustworthiness might have had something to do with their effect on how procedural justice is perceived, and the positive impact of explanations on accuracy might have had something to do with people’s belief that a system’s decisions are subject to correction.

We conjectured why there was a lack of strong evidence for the effect of E3 (a group-based explanation) across all attitudes. One possible reason is the nature of the decision. In the grant application scenario, the subject’s company submitted a proposal for funding for a new product or service, and the proposal was evaluated and rejected for an instance-specific reason; this is the type of information the subject might have been expecting. While case-based (E4) and counterfactual (E5) explanations served well in this regard, a group-based explanation (E3) would not have. For subjects seeking an instance-specific reason for the rejection, E3’s explanation regarding “the percentage of companies like yours

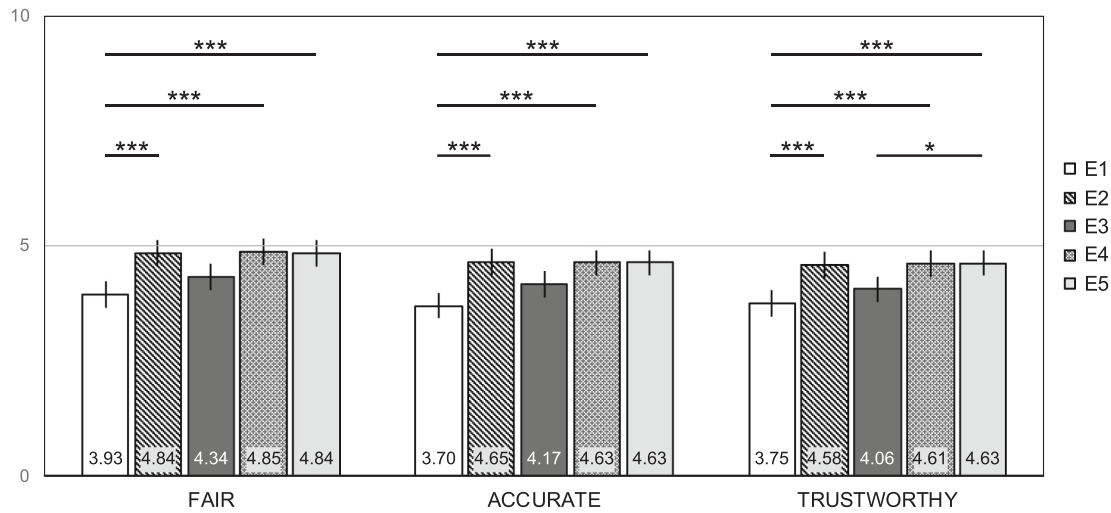


Fig. 3. Comparisons of means for perceived fairness, accuracy, and trustworthiness across the five experimental groups in the grant application study (Study 1). Note: The numbers indicate the means. The error bars indicate 95 % confidence intervals around the means. One (*) and three (***) asterisks indicate significance at the 0.05 and 0.001 levels, respectively.

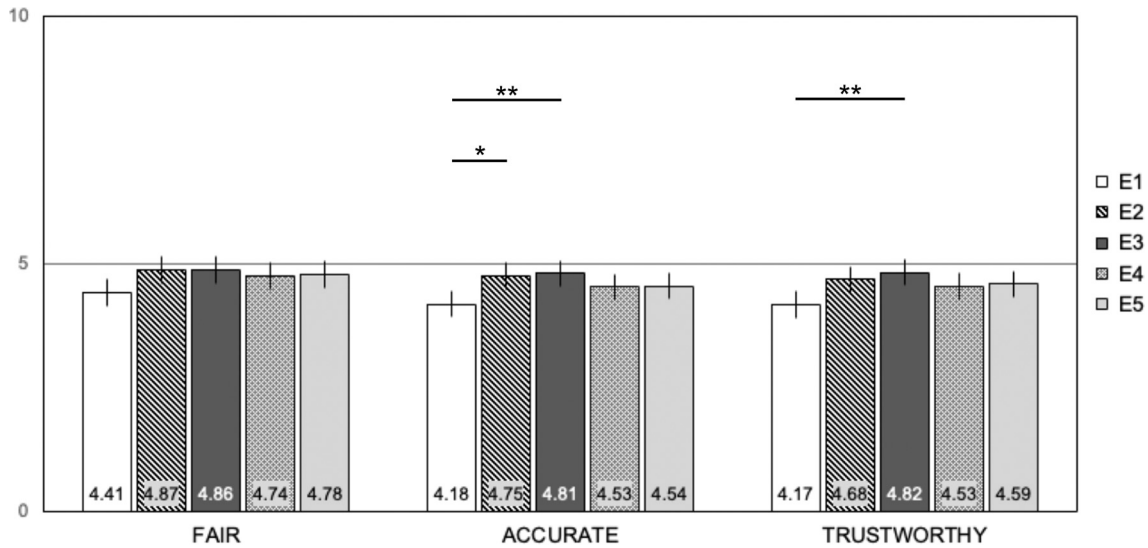


Fig. 4. Comparisons of means for perceived fairness, accuracy, and trustworthiness across the five experimental groups in the tax inspection study (Study 2). Note: The numbers indicate the means. The error bars indicate 95 % confidence intervals around the means. One (*) and two (**) asterisks indicate significance at the 0.05 and 0.01 levels, respectively.

... whose applications were accepted” might not have been the type of explanation they were looking for, and hence, the lack of a statistically significant difference between E1 (no explanation) and E3.

As for E2, input-based explanations are supposedly global and model-specific; however, E2 might have produced statistically significant results because, in this context of applying for a grant, some subjects who expected an instance-specific justification might have taken E2 as instance-specific, namely, as explaining how the inputs in *their* application contributed to the decision in *their* case, instead of understanding the explanation as applying to all decisions made by the system. This might be why the difference between E1 and E2 turned out to be statistically significant, while the difference between E1 and E3 did not.

In Study 2, interestingly, we did not find any statistically significant differences for fairness, unlike what we found in Study 1. This discrepancy between Studies 1 and 2 might have something to do with differences in the nature of the decisions. As noted above, subjects in Study 1 might have assumed that the decision was made for an instance-specific

reason, with the possibility of unfair treatment. As a result, they would have wanted to hear an instance-specific reason for why their proposal was rejected. On the other hand, in Study 2, regardless of how tax authorities actually select companies for an on-site tax inspection, many subjects might have had a de facto understanding that tax authorities are regularly conducting on-site tax inspections for a certain percentage of companies annually and that any company might be indiscriminately selected for an on-site inspection every once in a while.¹ As a result, the study subjects would not have felt the need for an explanation to justify the fairness of the decision – a possible reason for the insignificant

¹ This is what freee K.K., which sells tax-return software in Japan, explains on its website: Companies may be subject to a tax audit every four to five years, unless they have committed fraud in the past, which may make them subject to tax audits annually (freee K.K., 2023). However, actual selection may use a more targeted approach, according to the National Tax Office’s statement that it conducts on-site inspections of taxpayers with a high need for investigation and contact for the purpose of corrective action (National Tax Office, 2022).

results for fairness.

As for the statistically significant results we observed for accuracy and trustworthiness in Study 2, we earlier posited that any explanation, but especially a global one, would have a positive impact on perceived accuracy and trustworthiness. Our results are partially consistent with these hypotheses, in that explanations potentially perceived to be global – E2 and E3 for accuracy and E3 for trustworthiness – show statistically significant effects, while the other explanations do not. As for E2, we argued that it might not have been perceived as a global explanation in Study 1; however, in Study 2, where the nature of the decision is not so instance-specific, the opposite might have been the case. The subjects might have assumed that the report about the influence of the inputs applied indiscriminately to all companies and, thus, that the explanation was global rather than instance-specific, and hence the significant result.

There are some possible reasons for why the effects we found were small. First, the effects might have been more conspicuous if, instead of scenarios, the studies had involved real situations, such that the study subjects received actual decisions that negatively affected them, in which case, their reaction would have been stronger, and they would have been more eager to receive an explanation. Second, the effect sizes might have been larger if the explanations had contained actual information, such as dollar amounts and visuals; however, we deliberately avoided this to isolate the effects of the types of explanations. Third, the effects might have been more conspicuous if the decisions had negatively affected the subjects on a personal level, rather than primarily affecting their company. Fourth, it is possible that the effects were not conspicuous because some subjects did not fully understand the manipulated conditions – the types of information provided in the explanations. Lastly, the type of explanation might play only a small role in the variability of attitudes towards algorithmic decisions, which can be impacted by numerous factors.

On the last point above, this difficulty might have arisen especially because the subjects were unlikely to have seen actual explanations provided by Japanese authorities and because some of them might not have been familiar with what it takes to explain algorithmic decisions. In fact, some subjects in experimental groups E2 to E5 in both studies might not have thought that the types of explanations E2 to E5 provided sufficiently explicit information on why the decision was made. The results from the factual manipulation check, shown in Tables S-4 and S-5 in the Supplementary Material, suggest this possibility. Among those who failed the manipulation check (i.e., selected an explanation different from the one they had read in their scenario), we noticed that, aside from the “do not know” option, many of them disproportionately chose E1 (“you were given no explanation as to why this decision was made”). They might have thought E1 was the right answer without proceeding to read the statements for E2 to E5.

Direct comparisons between our studies and previous studies concerning the effects of explanations are difficult to make, due to discrepancies in research questions, designs, and contexts. Nevertheless, one close comparison we can make is between our findings and the finding in [Binns et al. \(2018\)](#) mentioned earlier, that the perceived fairness of the decision-making process in loan approvals was higher with a sensitivity-based (counterfactual) explanation than a case-based explanation. Contrary to this finding, our studies found no statistically significant difference in perceived fairness between counterfactual and case-based explanations. This contrast suggests the possibility that the effects of the types of explanations differ across decision contexts.

7. Conclusion

Amidst concern over potentially erroneous algorithmic decisions imposed by public authorities, it is important for members of the public to recognize that their authorities are making fair, accurate, and trustworthy decisions. We posited that this recognition depends in part on the type of explanation offered, because different kinds of information in the explanations would variably impact affectees’ assessment of the

equity and justice of the decision outcome, influencing both perceived fairness and trustworthiness. Moreover, the type of explanation influences perceived accuracy, because not all types of explanations would be seen to allow the monitoring of the underlying model so errors can be corrected. The results from our two experiments show that the type of explanation does matter, although the details of the results differ between them.

Our studies have limitations which can be addressed in future research. Despite the high internal validity built into the experimental design, our findings may lack external validity. Therefore, more studies can be conducted in other contexts and at other times with different samples to generate stronger evidence. Moreover, we tentatively explored the question of how the effects of different types of explanations differ across decision domains and why. For instance, we attributed the differences in the results for fairness between the two experiments to the different natures of the decisions and to whether or not the affectees wanted an instance-specific explanation. Since we did not produce empirical data to support our claim, future qualitative studies involving interviews with affectees might help uncover whether this is the case. Moreover, while our studies are limited to concerns about fairness, accuracy, and trustworthiness, future research could explore other aspects of algorithmic decisions, as perceived by affectees, that might deserve attention.

In regard to the significance of our studies, we contribute empirical insights to XAI research from societal and human-centric points of view, an approach that has been rather limited to date and a gap that needs to be addressed, especially in light of recent legal and regulatory moves for the right to obtain explanations for algorithmic decisions. More specifically, our studies offer findings in support of a general claim prior studies have put forward, that explanations influence public attitudes towards algorithmic decisions, whereby we reaffirm the possibility that the effects of different types of explanations differ across decision contexts. Conducting research in the public sector context was also meaningful, due to the paucity of research on the topic of XAI and explanations in the public sector. Moreover, we designed unprecedented experiments to isolate the effects of familiar types of explanations that differ with respect to the kind of information they present.

While further research is needed to generate strong and nuanced evidence, we can draw some tentative implications for practice. Our findings suggest that public authorities could consider providing some sort of explanation if they seek to promote affectees’ perceptions of the fairness, accuracy, and trustworthiness of their algorithmic decisions. In doing so, they are reminded that not all types of explanations are effective and that the effects are not the same for all kinds of decisions. While our studies do not make it possible to confirm which type of explanation is optimal for a specific decision that policy makers have in mind, they do demonstrate that the design of an explanation matters, that is, its type. This finding should prompt a nuanced inquiry into XAI discourse on how to design explanations from societal and human-centric perspectives.

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CRedit authorship contribution statement

Naomi Aoki: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Tomohiko Tatsumi:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Go Naruse:** Writing – review & editing, Methodology, Conceptualization. **Kentaro Maeda:** Writing – review & editing, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.giq.2024.101965>.

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